

Example: Let $X = \{a, b, c\}$. Can the class $\mathcal{B} = \{\{a, b\}, \{b, c\}\}$ be a base for a topology on X ?

Solution: No. Because if it were a base for a topology $\mathcal{T}$ on X then $\{a, b\}, \{b, c\} \in \mathcal{T}$. Because $\mathcal{T}$ is a top. on X

$\{a, b\} \cap \{b, c\} = \{b\} \in \mathcal{T}$ by (T2).

But then we would be able to express $\{b\}$ as a union of members of $\mathcal{B}$ which is impossible.

Now the following question arises;
Suppose we have an arbitrary class $\mathcal{B}$ of subsets of a nonempty set X .
When will the class $\mathcal{B}$ be a base for some topology on X ? Since X is open in every topology on X , clearly $X = \bigcup \mathcal{B} = \bigcup \{B \mid B \in \mathcal{B}\}$.

The following theorem gives necessary and sufficient

conditions for a class of subsets of X to be a base for some topology on X :

Theorem: Let $\mathcal{B}$ be a class of subsets of a nonempty set X . Then $\mathcal{B}$ is a base for some top. τ on $X \iff \mathcal{B}$ possesses the following properties:

$$(B1) \quad \bigcup \mathcal{B} = X$$

(B2) For every $B_1, B_2 \in \mathcal{B}$ & every $x \in B_1 \cap B_2$ there exists a $B_3 \in \mathcal{B}$ s.t. $x \in B_3 \subseteq B_1 \cap B_2$.

Proof:

$\Rightarrow$ (B1) Since $\mathcal{B}$ is a base for τ and $X \in \tau$, there exists a $\mathcal{B}' \subseteq \mathcal{B}$ s.t. $X = \bigcup \mathcal{B}'$.

But

$$X = \bigcup \mathcal{B}' \subseteq \bigcup \mathcal{B} \subseteq X, \text{ i.e.}$$

$$X = \bigcup \mathcal{B}.$$

(B2) Let $B_1, B_2 \in \mathcal{B}$ & $x \in B_1 \cap B_2$.

Since $\mathcal{B}$ is a base for $\mathcal{T}$,

$\mathcal{B} \subseteq \mathcal{T}$. Thus $B_1, B_2 \in \mathcal{T}$.

Because $\mathcal{T}$ is a top. on X

$B_1 \cap B_2 \in \mathcal{T}$ by (T_2) .

Now $B_1 \cap B_2 \in \mathcal{T}$ & $x \in B_1 \cap B_2$.

So there exists a $B_3 \in \mathcal{B}$ s.t.

$x \in B_3 \subseteq B_1 \cap B_2$ by the previous proposition.

$\Leftarrow$ Assume that $\mathcal{B}$ is a class of subsets of a nonempty set X satisfying (B1) & (B2). We shall show $\mathcal{B}$ is a base for some topology $\mathcal{T}$ on X . In particular we shall show that $\mathcal{B}$ is a base for

$$\begin{aligned} \mathcal{T} &= \{ \cup \mathcal{B}' \mid \mathcal{B}' \subseteq \mathcal{B} \} \\ &= \{ U \mid U \subseteq X \text{ and } x \in U \Rightarrow \exists B \in \mathcal{B} \text{ s.t. } x \in B \subseteq U \} \end{aligned}$$

But in this case we shall show that τ is a topology on X :

(T1) The empty class $\mathcal{A} = \{A_i \mid i \in \emptyset\}$ is a subclass of any class & so $\mathcal{A} \subseteq \mathcal{B}$. Moreover $\emptyset = \bigcup \mathcal{A}$. Hence $\emptyset \in \tau$.

By (B1) $X = \bigcup \mathcal{B}$ and $\mathcal{B} \subseteq \mathcal{B}$. Hence $X \in \tau$.

(T2) Let $U_1, U_2 \in \tau$.

Case 1 If $U_1 \cap U_2 = \emptyset$ then $U_1 \cap U_2 \in \tau$.

Case 2 Assume that $U_1 \cap U_2 \neq \emptyset$. Then there exists an $x \in X$ s.t. $x \in U_1 \cap U_2$.

$U_1 \in \tau, x \in U_1 \Rightarrow$ there exists a $B_1 \in \mathcal{B}$ s.t. $x \in B_1 \subseteq U_1$
and

$U_2 \in \tau, x \in U_2 \Rightarrow$ there exists a $B_2 \in \mathcal{B}$ s.t. $x \in B_2 \subseteq U_2$
by definition of τ .

Thus $x \in B_1 \cap B_2 \subseteq U_1 \cap U_2$. (1)

On the other hand because $B_1, B_2 \in \mathcal{B}$ & $x \in B_1 \cap B_2$ there is a $B_3 \in \mathcal{B}$ s.t. $x \in B_3 \subseteq B_1 \cap B_2$ by (B2). (2)

By (1) & (2)

$x \in B_3 \subseteq U_1 \cap U_2$ for some $B_3 \in \mathcal{B}$.

$\therefore U_1 \cap U_2 \in \tau$.

(T3) Let $U_i \in \mathcal{T}$ for $i \in I$

Case 1 If $U_i = \emptyset$ for all $i \in I$ $\bigcup_{i \in I} U_i = \emptyset$
and so $\bigcup_{i \in I} U_i \in \mathcal{T}$.

Case 2 Assume there is an $i_0 \in I$ s.t. $U_{i_0} \neq \emptyset$.
Then $x \in U_{i_0}$ for some $x \in X$.

$U_{i_0} \in \mathcal{T}$, $x \in U_{i_0} \Rightarrow$ there is a $B \in \mathcal{B}$ s.t.
 $\boxed{x \in B \subseteq U_{i_0}}$ ① by definition of $\mathcal{T}$.

On the other hand $U_{i_0} \subseteq \bigcup_{i \in I} U_i$ ②

By ① & ②

$x \in B \subseteq \bigcup_{i \in I} U_i$ for some $B \in \mathcal{B}$.

$\therefore \bigcup_{i \in I} U_i \in \mathcal{T}$.

OR

Let $U_i \in \mathcal{T}$ for $i \in I$. By definition of $\mathcal{T}$

there exists a $\mathcal{B}_j \subseteq \mathcal{B}$ s.t.

$$U_i = \bigcup_{j \in J} \mathcal{B}_j$$

Then

$$\bigcup_{i \in I} U_i = \bigcup_{i \in I} \left(\underbrace{\bigcup_{j \in J} \mathcal{B}_j}_{\mathcal{B}'} \right) = \bigcup \mathcal{B}'$$

$$\therefore \bigcup_{i \in I} U_i \in \mathcal{T}.$$

$\therefore \mathcal{T}$ is a topology on X .

Example: Let $X = \mathbb{R}$. Consider

the class $\mathcal{B}$ of all open intervals
 $\mathcal{B} = \{ (a, b) \mid a, b \in \mathbb{R}, a < b \}$. Then
 $\mathcal{B}$ forms a base for some top.
 $\mathcal{T}$ on $\mathbb{R}$. This topology is called
the usual (or the standard) top.
on $\mathbb{R}$. All open intervals are
open w.r.t. this topology.

Solution;

(B1) For every $x \in \mathbb{R}$, $x \in (x-1, x+1)$

Therefore

$$\mathbb{R} \subseteq \bigcup_{x \in \mathbb{R}} (x-1, x+1) \subseteq \bigcup \mathcal{B} \subseteq \mathbb{R}$$

$$\therefore \mathbb{R} = \bigcup \mathcal{B}$$

(B2) Let $B_1 = (a_1, b_1)$, $B_2 = (a_2, b_2) \in \mathcal{B}$

& let $x \in B_1 \cap B_2$.



Set $a_3 = \max \{a_1, a_2\}$ &

$b_3 = \min \{b_1, b_2\}$. Then

$$x \in (a_3, b_3) \subseteq B_1 \cap B_2 \quad ((a_3, b_3) \in \mathcal{B}).$$

$\mathcal{B}$ is a base for $\mathcal{T}$.

Example; Let $X = \mathbb{R}$. Let

$\mathcal{D}$ and $\mathcal{T}$ be the discrete top.
& the usual top. on $\mathbb{R}$, resp.

Show that $\mathcal{T} \subsetneq \mathcal{D}$.

Solution:

First note that $\mathcal{T} \subseteq \mathcal{D}$.

Since $\mathcal{D}$ is the discrete top.
on $\mathbb{R}$, $\{x\} \in \mathcal{D}$ for every
 $x \in \mathbb{R}$. But $\{x\}$ is not $\mathcal{T}$ -open.
Otherwise we would be able to
express $\{x\}$ as a union of
open intervals in $\mathbb{R}$ which is
impossible.

$$\therefore \mathcal{C} \subsetneq \mathcal{D}$$

Example: Let $X = \mathbb{R}$ &

$\mathcal{B} = \{(a, b] \mid a, b \in \mathbb{R}, a < b\}$. Show that

$\mathcal{B}$ is a base for some topology τ on $\mathbb{R}$.

τ is called the upper limit topology on $\mathbb{R}$.

Solⁿ We shall show that $\mathcal{B}$ satisfies (B1) & (B2).

(B1) Let $x \in \mathbb{R}$. Then $x \in (x-1, x]$.

Thus

$$\mathbb{R} \subseteq \bigcup_{x \in \mathbb{R}} (x-1, x] = \bigcup \mathcal{B}' \subseteq \bigcup \mathcal{B} \subseteq \mathbb{R}$$

where $\mathcal{B}' = \{(x-1, x] \mid x \in \mathbb{R}\}$.

Therefore

$$\bigcup \mathcal{B} = \mathbb{R}.$$

(B2) Let $B_1 = (a_1, b_1]$, $B_2 = (a_2, b_2] \in \mathcal{B}$
& $x \in (a_1, b_1] \cap (a_2, b_2] = B_1 \cap B_2$.



Set $a_3 = \max \{a_1, a_2\}$ & $b_3 = \min \{b_1, b_2\}$

Let $B_3 = (a_3, b_3]$. Then

$$x \in B_3 \subseteq B_1 \cap B_2.$$

$\therefore \mathcal{B}$ is a base for some
top. τ on $\mathbb{R}$.

Example: Let $X = \mathbb{R}^2$ & let $\mathcal{B}$
be the class of all open discs in $\mathbb{R}^2$.
Show that $\mathcal{B}$ is a base for some
top. τ on $\mathbb{R}^2$. (τ is the usual
top. on $\mathbb{R}^2$).

Solⁿ As in the previous example
we shall show that $\mathcal{B}$ possesses
(B1) & (B2).

(B1) Let $x = (x_1, x_2) \in \mathbb{R}^2$. Then
 $x \in B(x, 1)$. Hence if
 $\mathcal{B}' = \{B(x, 1) \mid x \in \mathbb{R}^2\}$ we have